Modeling Bimanual Perceptual Interactions in Macaque Monkeys

Bayesian Hierarchical Modeling

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Outline

Task and Data
Model A: Sessions Pooled
- Model Structure
- Synthetic Data
- Real Data

Model B: Sessions Hierarchical
- Model Structure
- Synthetic Data
- Real Data

Model C: Distractor Amplitude as a Covariate
- Model Structure
- Synthetic Data
- Real Data

Model Comparison and Recovery
- Predictive Performance
- Model Recovery: B vs A
- Model C vs Model B

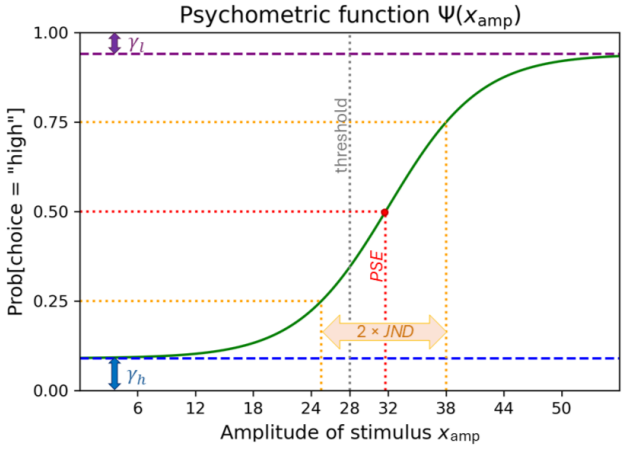
Summary

The task: 2AFC categorization of vibration amplitude

- stimulus $x_{\text{stim}} \in \{6, 12, 18, 24, 32, 38, 44, 50\}\ \mu m$
  - administered to one finger of one hand (left or right)
- distractor $x_{\text{dist}} \in \{0, x_{\text{d}}\}\ \mu m$
  - $x_{\text{d}} \in [6, 32]\ \mu m$, value fixed for each session
  - administered to the other (non-stimulus) hand
- if present, distractor administered simultaneously with the stimulus
- monkey categorizes the stimulus as either "high" or "low" relative to a trained boundary of 28 μm
  - "low" response: correct for $x_{\text{stim}} < 28\ \mu m$
  - "high" response: correct for $x_{\text{stim}} > 28\ \mu m$
- correct responses are rewarded with water or juice

Psychometric function: response $r \sim \text{Bernoulli}(p = \psi)$

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}})$$

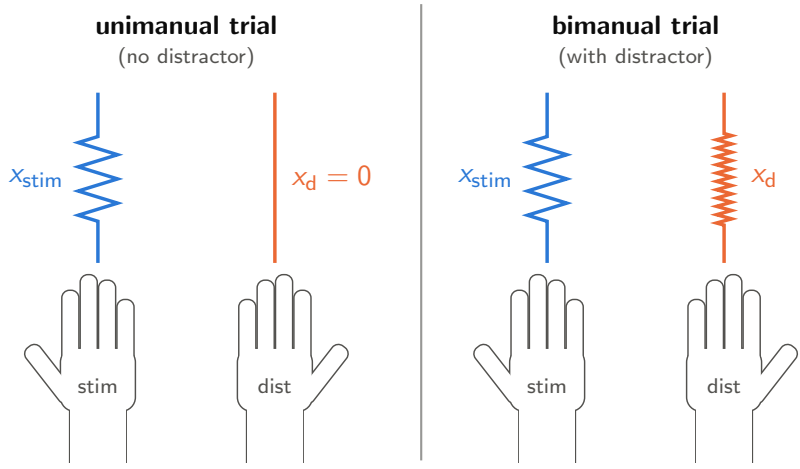


Parameters

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}}), \qquad \vec{\theta} = (\beta_0, \beta_1, \gamma_h, \gamma_l)$$

- γ_h: lapse rate for "high"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"high"})$, even to the weakest stimuli
  - lower (left) asymptote of ψ
- γ_l: lapse rate for "low"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"low"})$, even to the strongest stimuli
  - $1 - \gamma_l$ = upper (right) asymptote of ψ
- logit^{-1}: the logistic sigmoid, $\text{logit}^{-1}(z) = \dfrac{1}{1 + \exp(-z)}$
- PSE: stimulus where responses of "high" and "low" are equally likely
  - x_{stim} such that $\psi(x_{\text{stim}}) = 0.5$
  - represents bias
- JND: how much the stimulus needs to change to be noticed half the time
  - half the distance between $\psi(x_{\text{stim}}) = 0.25$ and 0.75
  - represents sensitivity

The task: one trial



Modeling bimanual interactions

- stimulus $x_{\text{stim}} \in \{6, 12, 18, 24, 32, 38, 44, 50\}\ \mu m$
  - administered to one finger of one hand (left or right)
- distractor $x_{\text{dist}} \in \{0, x_{\text{d}}\}\ \mu m$
  - $x_{\text{d}} \in [6, 32]\ \mu m$, value fixed for each session
  - administered to the other (non-stimulus) hand
- if present, distractor administered simultaneously with the stimulus
- monkey categorizes the stimulus as either "high" or "low" relative to a trained boundary of 28 μm
  - "low" response: correct for $x_{\text{stim}} < 28\ \mu m$
  - "high" response: correct for $x_{\text{stim}} > 28\ \mu m$
- correct responses are rewarded with water or juice

The data

- One monkey (Sirius), 12,937 trials.
- Collected over 42 sessions, averaging approximately 308 trials per session.
- 8 stimulus amplitudes, 6–50 μm; 4 groups: hand $\times$ (unimanual, bimanual).
- Each session uses a single distractor amplitude (6–32 μm) on its bimanual trials.

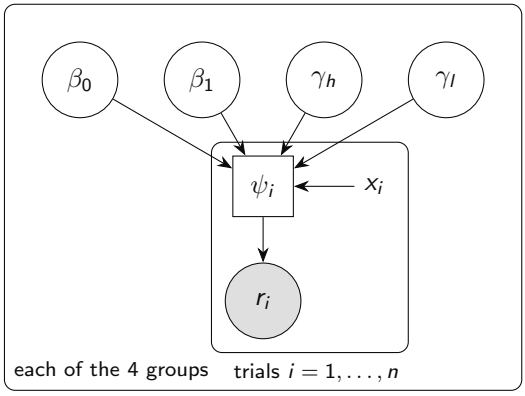
	left hand		right hand		
x_{stim} (μm)	unimanual	bimanual	unimanual	bimanual	total
6	408	404	404	402	1,618
12	407	400	408	400	1,615
18	402	400	403	405	1,610
24	402	405	402	406	1,615
32	404	405	405	407	1,621
38	403	408	404	408	1,623
44	406	403	404	405	1,618
50	405	405	406	401	1,617
total	3,237	3,230	3,236	3,234	12,937

Three models

Model	Sessions	Distractor
A	pooled: one curve per group	uni and bi fit separately
B	hierarchical: one curve per group and session	uni and bi fit separately
C	hierarchical: one curve per hand and session	covariate: shift $\beta_2 x_{\text{dist}}$

For each model: convergence study on synthetic data, then the fit to the real data.

Model A: graphical model



- circles: random variables
- shaded: observed
- rectangles: deterministic
- x_i: fixed covariate
- outer plate: the same model, fit independently to each group

Model A: likelihood

For each group (left/right $\times$ unimanual/bimanual), fit independently to its $n \approx 3{,}230$ trials from all 42 sessions pooled:

Likelihood: probability of obtaining responses $\vec{r}$ given parameters $\vec{\theta} = (\beta_0, \beta_1, \gamma_h, \gamma_l)$
$r_i \sim \textbf{Bernoulli}[\psi(x_i; \vec{\theta})]$ ($r_i = 1$ for "high", x_i standardized), so:

$$\mathbf{P}[r_i = z] = \psi(x_i; \vec{\theta})^z \left(1 - \psi(x_i; \vec{\theta})\right)^{1-z}, \qquad z \in \{0, 1\}$$

$$\begin{aligned} L(\vec{\theta}; \vec{r}) &= \prod_{i=1}^{n} \mathbf{P}[r_i \mid \vec{\theta}] \\ &= \prod_{i=1}^{n} \psi(x_i; \vec{\theta})^{r_i} \left(1 - \psi(x_i; \vec{\theta})\right)^{1-r_i} \end{aligned}$$

Model A: priors

The same priors in every group:

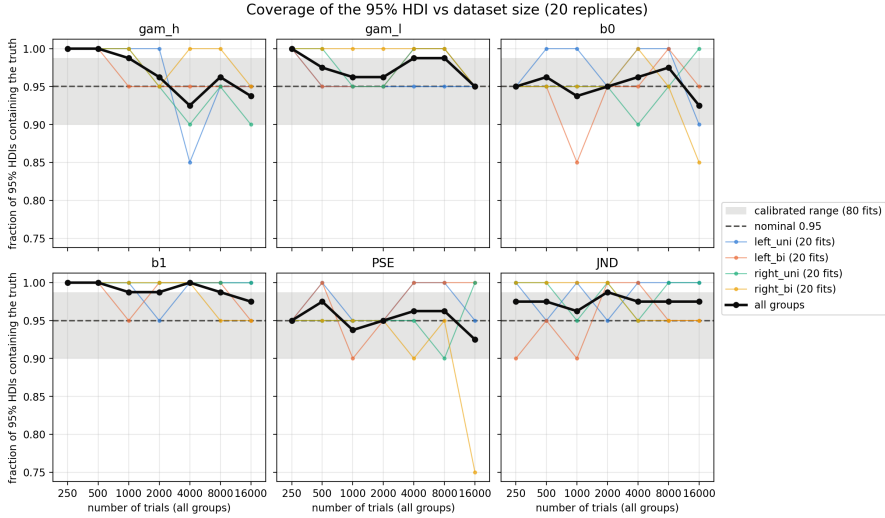
$$\begin{aligned}
\beta_0 &\sim \mathcal{N}(0,\ 6) \\
\beta_1 &\sim \mathcal{N}(4,\ 2) \\
\gamma_h &= 0.25\, w_h, \qquad & w_h &\sim \text{Beta}(1.22,\ 4.89) \\
\gamma_l &= 0.25\, w_l, \qquad & w_l &\sim \text{Beta}(1.22,\ 4.89)
\end{aligned}$$

- β_1 should be positive for the psychometric function to have a positive slope
- γ's have support between 0 and 0.25, with prior mean of 0.05 and 95% range $[0.003,\ 0.141]$

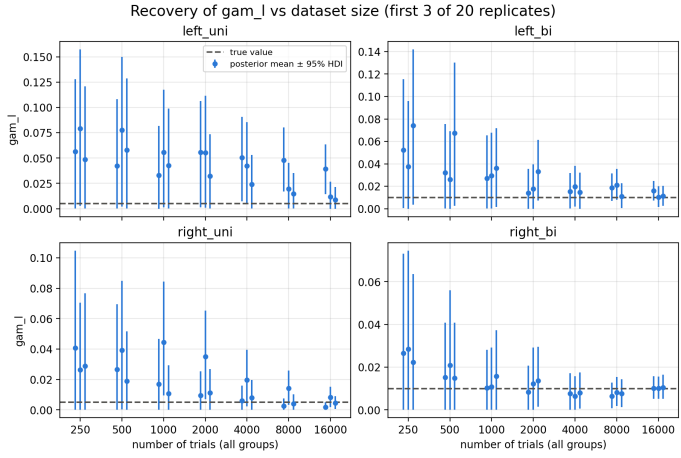
Model A on synthetic data: the study

- One fixed set of true parameters $(\beta_0, \beta_1, \gamma_h, \gamma_l)$ per group
- Trials drawn from the real design (the real mix of stimulus levels and groups), responses simulated from the true curves
- 7 sizes: 250 to 16,000 trials; the real data (12,937) lie between the two largest
- 20 replicates of the whole sweep: 140 fits, 80 per size and parameter
- All fits sampled well: $\hat{R} \leq 1.006$, bulk ESS $\geq$ 1,554

Model A on synthetic data: the 95% HDIs are calibrated



Model A on synthetic data: γ_l is pulled upward



True γ_l (0.005, 0.01) sits in the prior's lower tail, and the data hold only 3–7 such lapses: the estimates are 1 posterior sd too high at small sizes, 0.36 sd at 16,000.

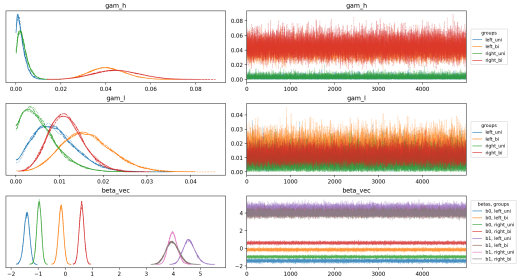
Model A on synthetic data: what is recoverable

At the size of the real data (about 3,230 trials per group):

- **Recoverable:** PSE, JND, β_0, γ_h: unbiased, calibrated HDIs
  - posterior sd about 0.3 μm for the PSE, 0.2 μm for the JND
- **Recoverable, HDIs a little wide:** β_1 (coverage 0.98–0.99)
- **Not reliably recoverable:** small γ_l
  - biased upward by 0.4–0.5 sd; by about 1 sd when the curve never reaches its ceiling (left unimanual), where γ_l trades off with the slope
  - HDIs still cover the truth, but the posterior mean should not be read literally
- **Caveats:** one set of true values; data simulated with one curve per group, so this says nothing about variation between sessions

Model A on the real data: sampling diagnostics

- NUTS, 4 chains $\times$ 5,000 draws after 1,000 tuning (2.4 min)
- No divergences; tree depth never above 5
- All 40 parameters: $\hat{R} = 1.000$, bulk ESS $\geq$ 15,213, tail ESS $\geq$ 8,802



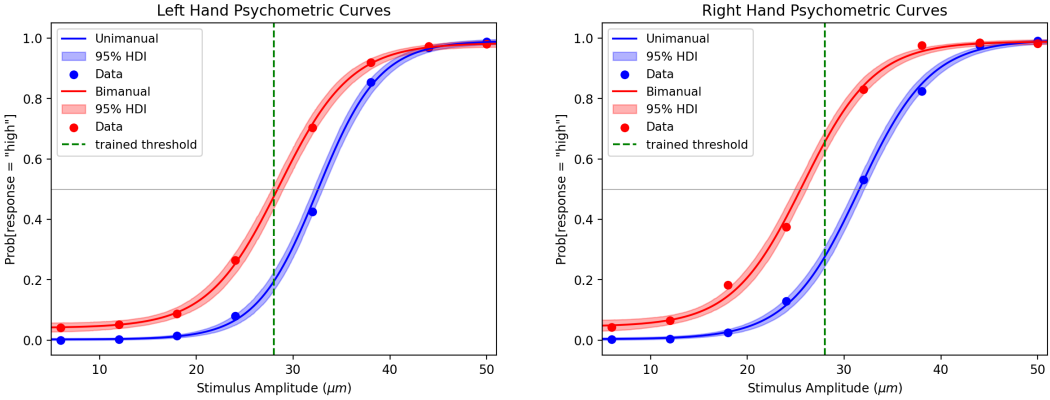
Model A on the real data: group estimates

	PSE (μm)	JND (μm)	γ_h	γ_l
left unimanual	32.58 [32.10, 33.04]	3.56 [3.25, 3.89]	0.002 [0.000, 0.006]	0.009 [0.000, 0.018]
left bimanual	28.38 [27.79, 28.98]	4.39 [3.93, 4.83]	0.041 [0.025, 0.056]	0.016 [0.005, 0.027]
right unimanual	31.57 [31.05, 32.06]	4.09 [3.77, 4.41]	0.003 [0.000, 0.008]	0.006 [0.000, 0.013]
right bimanual	25.50 [24.89, 26.08]	4.36 [3.93, 4.81]	0.045 [0.025, 0.064]	0.012 [0.004, 0.020]

Posterior mean [95% HDI], 12,937 trials.

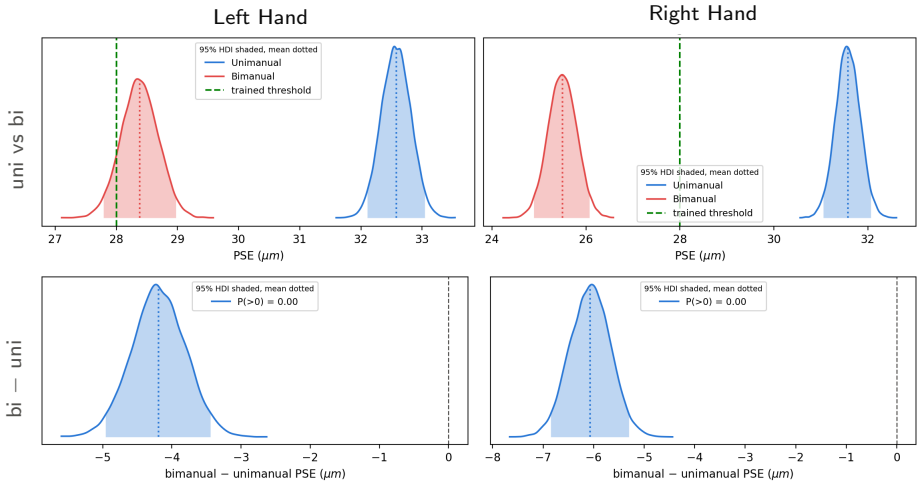
- Without the distractor the PSE lies 3.6–4.6 μm above the trained 28 μm
- With it the PSE moves to the threshold (left) or below it (right)
- γ_l estimates lean high (see the synthetic study)

Model A on the real data: fitted curves



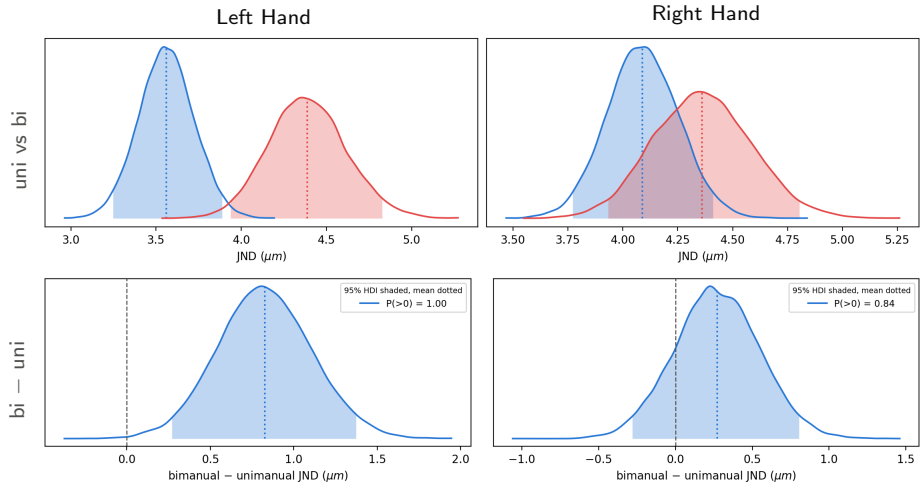
Unimanual in blue, bimanual in red, with the observed proportions pooled over sessions; dashed: the trained threshold of 28 μm.

Model A on the real data: effect of the distractor on PSE

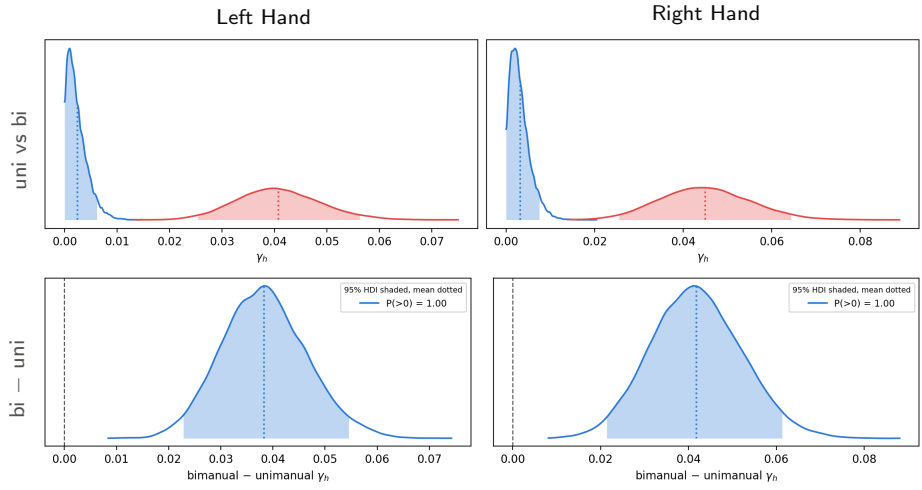


Unimanual Bimanual 95% HDI shaded, mean dotted; green dashed: trained threshold

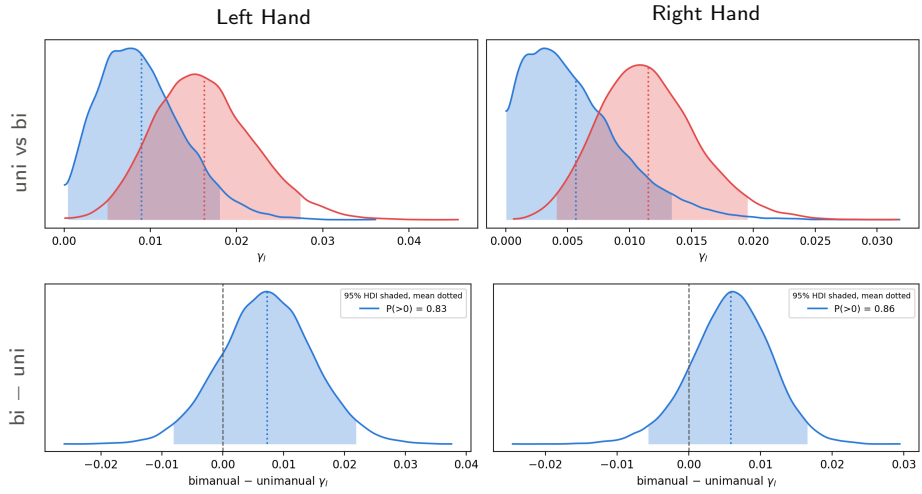
Model A on the real data: effect of the distractor on JND



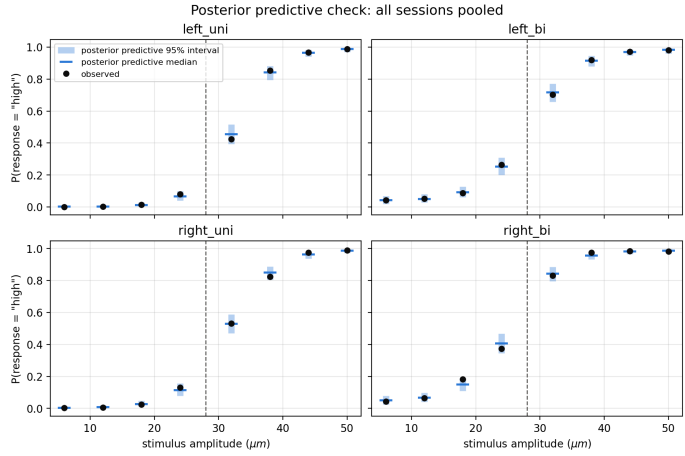
Model A on the real data: effect of the distractor on γ_h



Model A on the real data: effect of the distractor on γ_l



Model A on the real data: the pooled data are reproduced



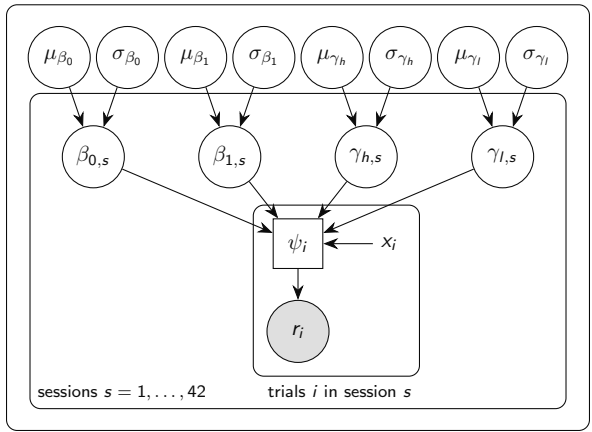
Pooled over sessions, all 32 group $\times$ amplitude proportions lie inside their 95% posterior predictive intervals.

Model A: conclusions

- Reliable fit; the pooled data are reproduced at every amplitude ($\text{elpd}_{\text{loo}} = -3344.0$, SE 64.5)
- Effect of the distractor (bimanual − unimanual), posterior mean [95% HDI]:
  - **PSE:** lowered, by -4.19 [-4.96, -3.44] μm (left) and -6.07 [-6.85, -5.30] μm (right)
  - **JND:** raised for the left hand, $+0.83$ [0.27, 1.37] μm; unclear for the right, $+0.27$ [-0.29, 0.80] μm
  - γ_h**:** raised, by $+0.038$ [0.023, 0.055] (left) and $+0.042$ [0.021, 0.061] (right)
  - γ_l**:** no clear effect, $+0.007$ [-0.008, 0.022] (left) and $+0.006$ [-0.006, 0.017] (right)

$\Rightarrow$ A curve for every session: Model B

Model B: graphical model



Circles: random variables (shaded: observed); rectangle: deterministic; x_i: fixed covariate. The same model is fit independently to each group.

Model B: likelihood

For each group (left/right $\times$ unimanual/bimanual), fit independently to its $n \approx 3{,}230$ trials, now with a parameter set for every session s:

Likelihood: probability of obtaining responses $\vec{r}$ given parameters $\vec{\theta} = (\vec{\theta}_s)_{s=1}^{42}$,
$\vec{\theta}_s = (\beta_{0,s}, \beta_{1,s}, \gamma_{h,s}, \gamma_{l,s})$
$r_i \sim \textbf{Bernoulli}[\psi(x_i; \vec{\theta}_{s[i]})]$ ($s[i]$: the session of trial i), so:

$$\mathbf{P}[r_i = z] = \psi(x_i; \vec{\theta}_{s[i]})^z \left(1 - \psi(x_i; \vec{\theta}_{s[i]})\right)^{1-z}, \qquad z \in \{0, 1\}$$

$$\begin{aligned} L(\vec{\theta}; \vec{r}) &= \prod_{i=1}^{n} \mathbf{P}[r_i \mid \vec{\theta}_{s[i]}] \\ &= \prod_{s=1}^{42} \prod_{i:\, s[i]=s} \psi(x_i; \vec{\theta}_s)^{r_i} \left(1 - \psi(x_i; \vec{\theta}_s)\right)^{1-r_i} \end{aligned}$$

Model B: priors

Session parameters drawn from the group's population (non-centered in the code):

$$\beta_{0,s} \sim \mathcal{N}(\mu_{\beta_0},\ \sigma_{\beta_0}), \qquad \gamma_{h,s} = 0.25\, w_{h,s}, \quad w_{h,s} \sim \text{LogitNormal}(\mu_{\gamma_h},\ \sigma_{\gamma_h})$$
$$\beta_{1,s} \sim \mathcal{N}(\mu_{\beta_1},\ \sigma_{\beta_1}), \qquad \gamma_{l,s} = 0.25\, w_{l,s}, \quad w_{l,s} \sim \text{LogitNormal}(\mu_{\gamma_l},\ \sigma_{\gamma_l})$$

Hyperpriors, the same in every group:

$$\mu_{\beta_0} \sim \mathcal{N}(0,\ 2), \qquad \sigma_{\beta_0} \sim \text{HalfNormal}(2),$$
$$\mu_{\beta_1} \sim \text{LogNormal}(\log 4,\ 0.5), \qquad \sigma_{\beta_1} \sim \text{Exponential}(1),$$
$$\mu_{\gamma_h},\ \mu_{\gamma_l} \sim \mathcal{N}(-3,\ 1.5), \qquad \sigma_{\gamma_h},\ \sigma_{\gamma_l} \sim \text{HalfNormal}(0.5)$$

- Chosen from the design, checked by prior predictive simulation, not fit to the data
- μ_{β_0}: typical PSE within 13–43 μm with 90% probability
- $\mu_{\beta_1} > 0$: median 4, 95% in [1.5, 10.6]; σ_{β_1} keeps 98% of session slopes positive
- Typical lapse rate: median 0.012, 95% in [0.0007, 0.12]

Model B on synthetic data: the study

- True hyperparameters = posterior means of Model B's fit to the real data (next slide), so the synthetic data resemble the real data; session parameters drawn from them
- Two sweeps, each run twice with new data:
  - sessions: 5, 10, 20, 40, 80 sessions of 308 trials (tests the hyperparameters)
  - trials: 42 sessions of 20, 40, 80, 160, 320 trials (tests the session parameters)
- The real data (42 sessions of about 308 trials) are closest to the 40-session and 320-trial fits
- All 20 fits sampled well: no divergences, $\hat{R} \leq 1.011$, bulk ESS ≥ 670

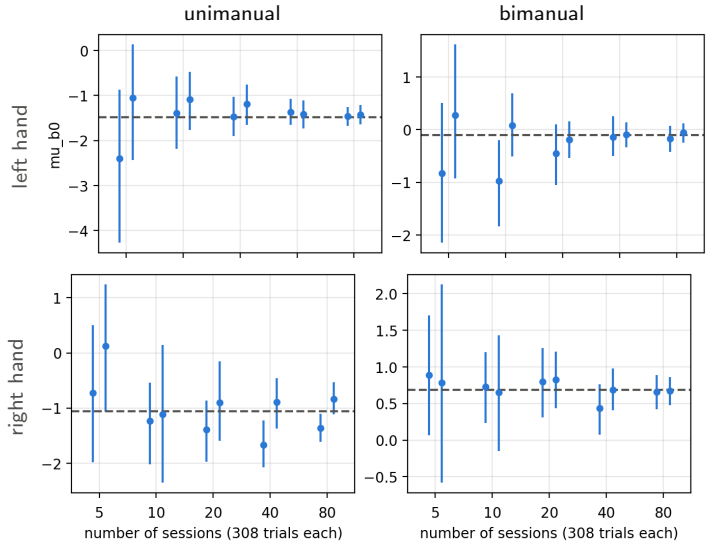
Model B on synthetic data: the true values

Posterior means of the real-data fit (lapse-rate hyperparameters on the logit scale):

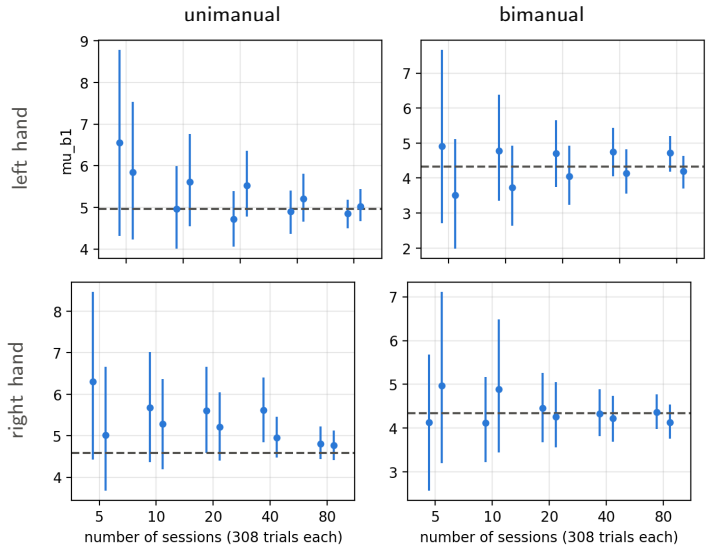
	μ_{β_0}	μ_{β_1}	μ_{γ_h}	μ_{γ_l}	σ_{β_0}	σ_{β_1}	σ_{γ_h}	σ_{γ_l}
left unimanual	−1.49	4.97	−5.20	−3.44	0.76	0.29	0.36	0.37
left bimanual	−0.10	4.32	−2.35	−3.44	0.77	1.27	0.59	0.39
right unimanual	−1.06	4.58	−4.73	−3.72	1.01	0.17	0.37	0.38
right bimanual	0.69	4.34	−1.82	−3.23	0.76	0.62	0.36	0.43

- Group slopes 4.3–5.0, and slopes that vary between sessions (left bimanual most)
- Typical γ_h near 0 on unimanual trials (0.001–0.002), 0.02–0.04 on bimanual
- Replaces an earlier study with hand-picked values (slopes 3 and 6, sessions with almost equal slopes and lapse rates) that did not resemble the real data

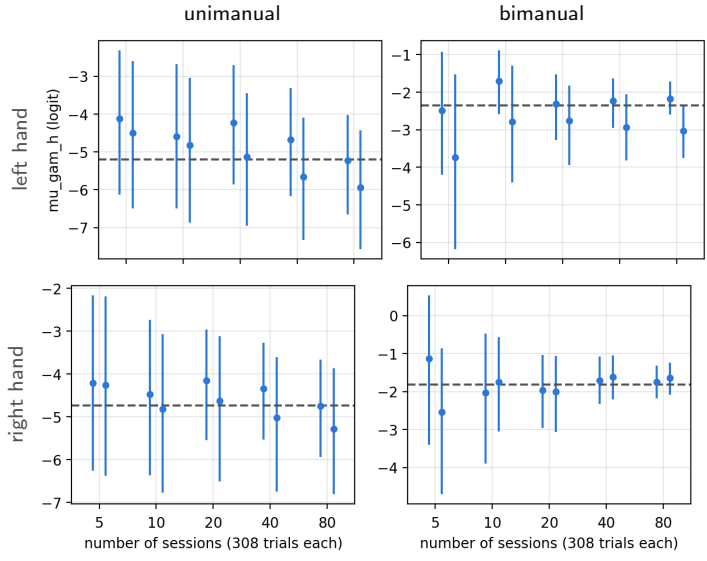
Model B on synthetic data: μ_{β_0} vs number of sessions



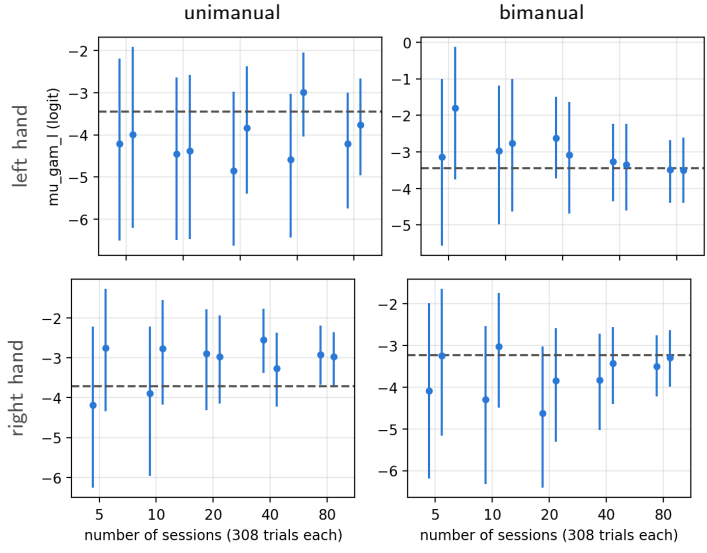
Model B on synthetic data: μ_{β_1} vs number of sessions



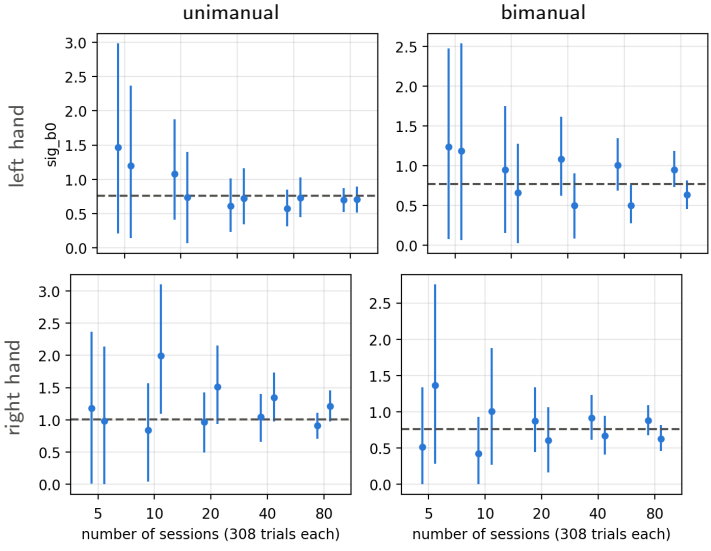
Model B on synthetic data: μ_{γ_h} vs number of sessions



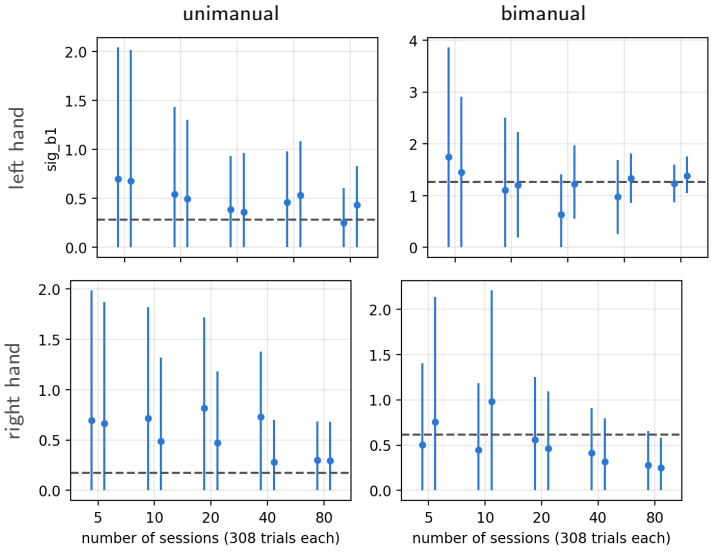
Model B on synthetic data: μ_{γ_I} vs number of sessions



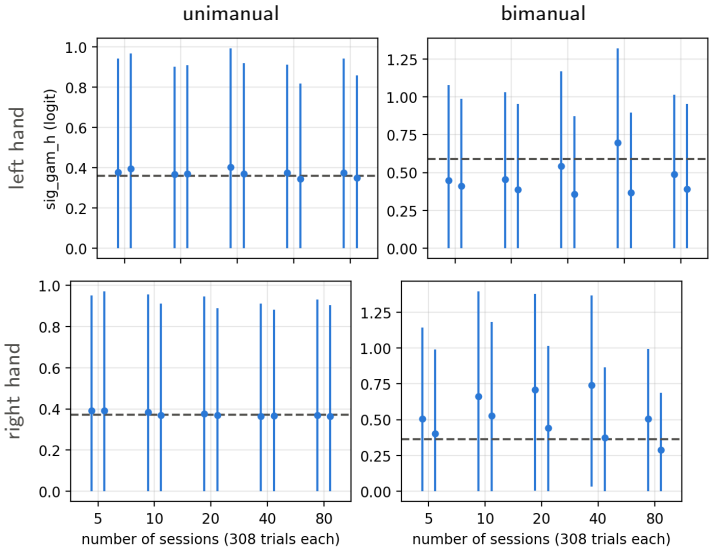
Model B on synthetic data: σ_{β_0} vs number of sessions



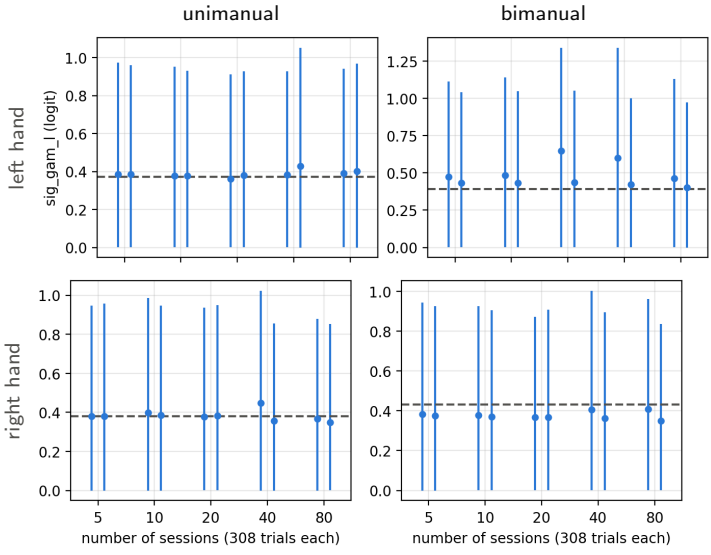
Model B on synthetic data: σ_{β_1} vs number of sessions



Model B on synthetic data: σ_{γ_h} vs number of sessions



Model B on synthetic data: σ_{γ_I} vs number of sessions



Model B on synthetic data: session-level recovery

At 320 trials per session (42 sessions, both replicates): RMSE of the posterior mean / mean 95% HDI width / coverage

	left unimanual	left bimanual	right unimanual	right bimanual
PSE	0.08 / 0.33 / 0.95	0.11 / 0.42 / 0.98	0.11 / 0.38 / 0.92	0.11 / 0.40 / 0.92
JND	0.017 / 0.07 / 0.98	0.095 / 0.28 / 0.93	0.013 / 0.08 / 1.00	0.039 / 0.16 / 0.98
β_0	0.45 / 1.83 / 0.96	0.40 / 1.47 / 0.92	0.49 / 1.76 / 0.90	0.46 / 1.75 / 0.92
β_1	0.44 / 1.88 / 0.99	1.02 / 2.90 / 0.87	0.23 / 1.39 / 1.00	0.60 / 2.43 / 0.94
γ_h	0.001 / 0.005 / 1.00	0.018 / 0.023 / 0.57	0.002 / 0.008 / 1.00	0.018 / 0.054 / 0.86
γ_l	0.009 / 0.031 / 1.00	0.008 / 0.026 / 0.99	0.003 / 0.012 / 0.96	0.006 / 0.030 / 1.00

PSE and JND in standardized units (1 unit = 14.6 μm).

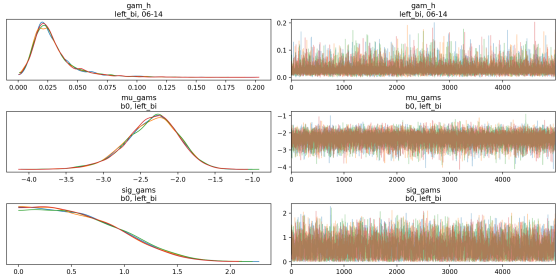
Model B on synthetic data: what is recoverable

At the size of the real data (42 sessions of about 308 trials), at values like the real ones:

- **Recoverable:** per-session PSE (RMSE about 1.5 μm) and JND; the group means μ_{β_0} and μ_{β_1}; the bias spread σ_{β_0}; the slope spread σ_{β_1} when it is large (left bimanual)
- **Weakly determined:** the lapse-rate means, especially μ_{γ_h} for unimanual trials (γ_h near 0); smaller slope spreads, which cannot be told from 0
- **Biased in some datasets:** μ_{γ_l} for right unimanual (too high); γ_h for left bimanual (too low) and γ_l for left unimanual (too high), where the curve does not reach that asymptote
- **Overconfident:** session lapse rates. A miss in the group's γ_h carries over to every session: bimanual session γ_h HDIs cover the truth in only 57% (left) and 86% (right) of sessions
- **Not identified:** σ_{γ_h}, σ_{γ_l}; their posteriors reflect the prior

Model B on the real data: sampling diagnostics

- NUTS, 4 chains $\times$ 5,000 draws after 1,000 tuning (31 min)
- No divergences; tree depth never above 6
- All but 4 of the 2,080 parameters: $\hat{R} = 1.00$, bulk ESS $\geq$ 2,210, tail ESS $\geq$ 4,527
- The 4 exceptions (raw `z_sig_b0`, $\hat{R}$ up to 1.73) are harmless: $\sigma_{\beta_0} = |2z|$, so chains differ only in the sign of z; σ_{β_0} itself has $\hat{R} = 1.00$
- Below: left bimanual γ_h (06-14) and its hyperparameters; σ_{γ_h} piles up at 0 but mixes well



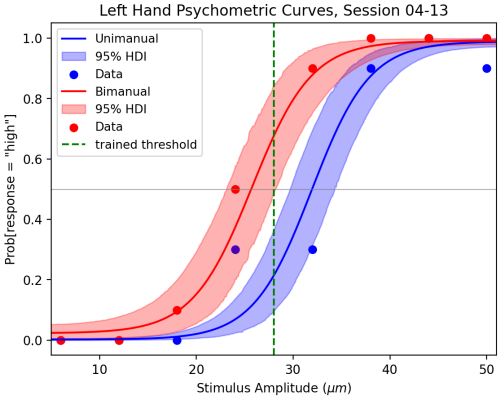
Model B on the real data: a typical session

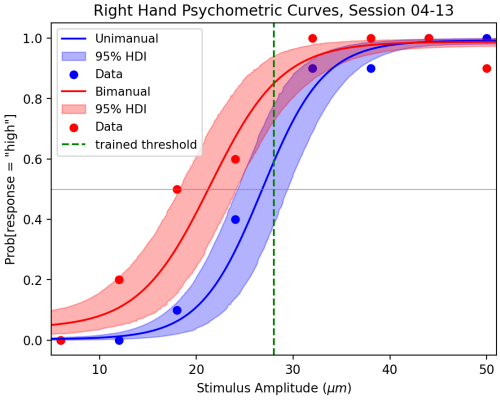
	PSE (μm)	JND (μm)	γ_h	γ_l
left unimanual	32.4 [31.6, 33.3]	3.29 [2.95, 3.63]	0.002 [0.000, 0.005]	0.009 [0.000, 0.019]
left bimanual	28.2 [27.2, 29.2]	3.89 [3.34, 4.46]	0.023 [0.009, 0.037]	0.009 [0.001, 0.017]
right unimanual	31.4 [30.3, 32.5]	3.56 [3.23, 3.89]	0.003 [0.000, 0.006]	0.007 [0.000, 0.015]
right bimanual	25.5 [24.5, 26.5]	3.95 [3.48, 4.44]	0.036 [0.017, 0.056]	0.010 [0.003, 0.018]

Quantities at the group-level means, posterior mean [95% HDI].

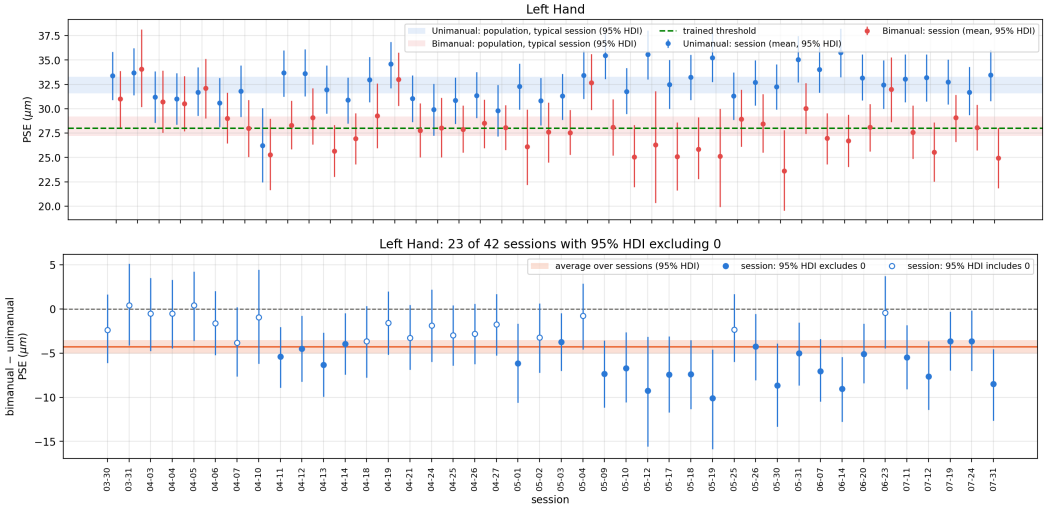
- The sessions differ in bias ($\sigma_{\beta_0} = 0.76$–1.01); left bimanual also in slope ($\sigma_{\beta_1} = 1.27$)
- The between-session lapse sds sit near their prior (not identified)

Model B on the real data: fitted curves in one session (04-13)

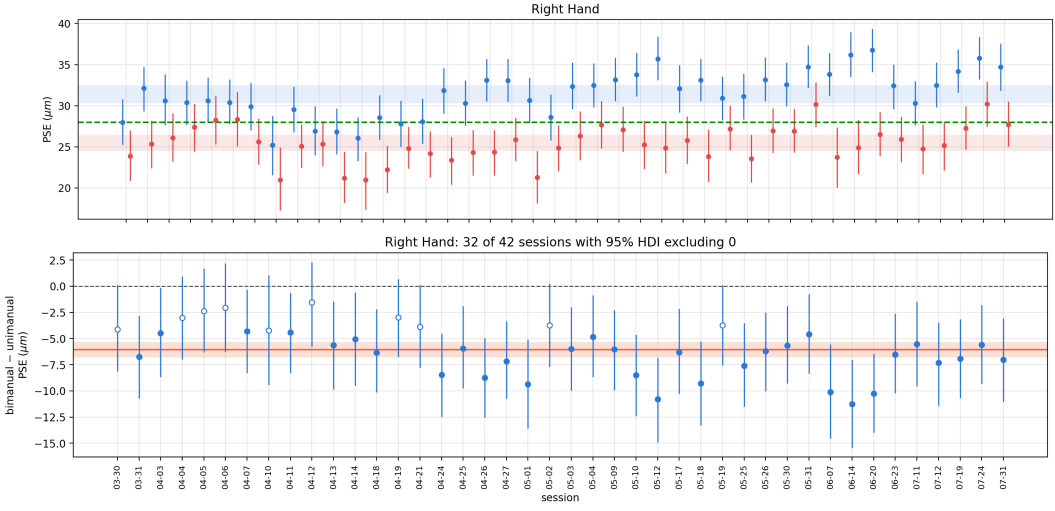




Model B on the real data: PSE, left hand



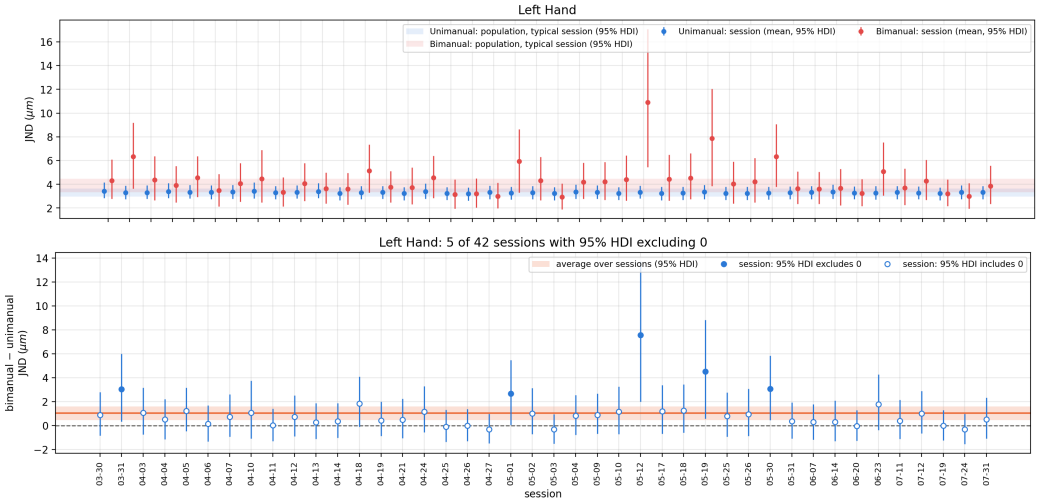
Model B on the real data: PSE, right hand



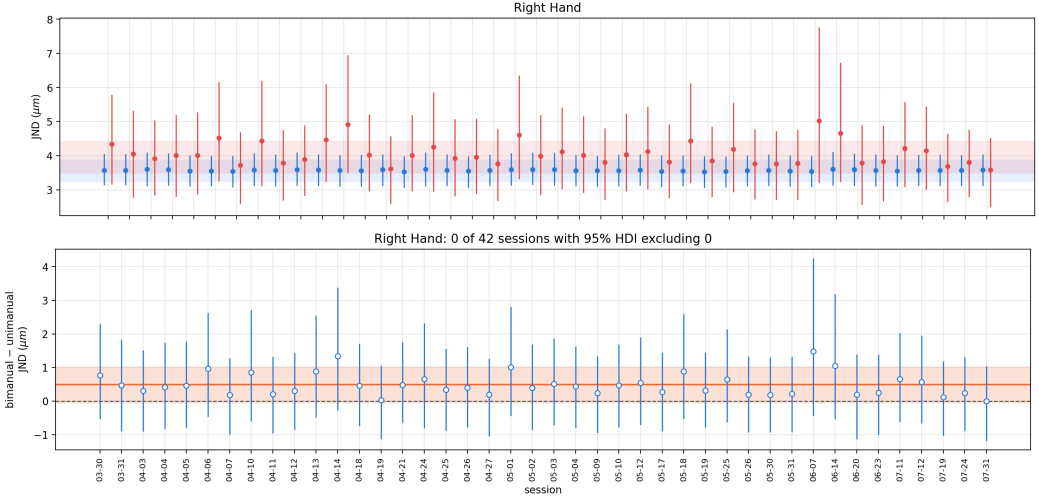
● unimanual, ● bimanual: session mean and 95% HDI; bands: typical session; green dashed: trained threshold

bi − uni: filled if the session's 95% HDI excludes 0; orange band: average over sessions

Model B on the real data: JND, left hand



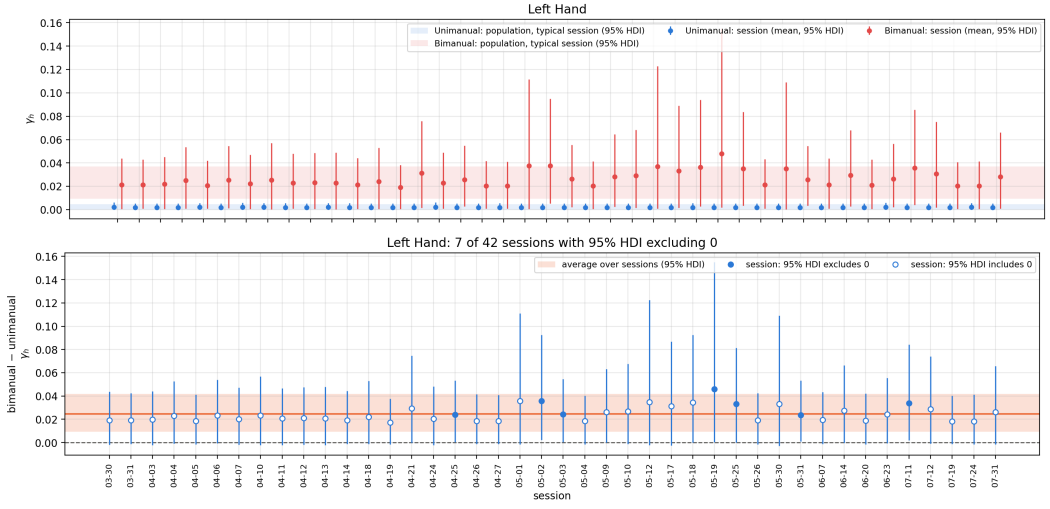
Model B on the real data: JND, right hand



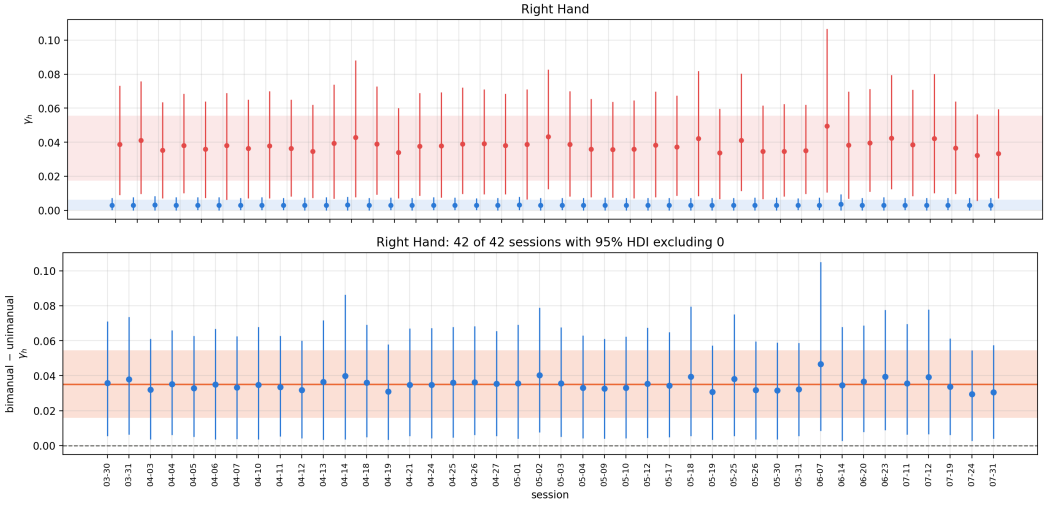
● unimanual, ● bimanual: session mean and 95% HDI; bands: typical session

bi − uni: filled if the session's 95% HDI excludes 0; orange band: average over sessions

Model B on the real data: γ_h, left hand



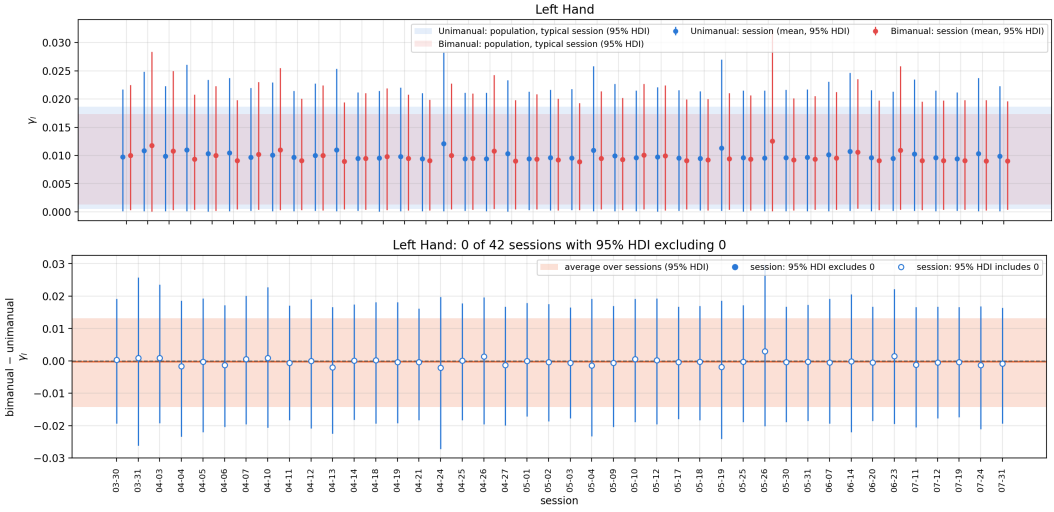
Model B on the real data: γ_h, right hand



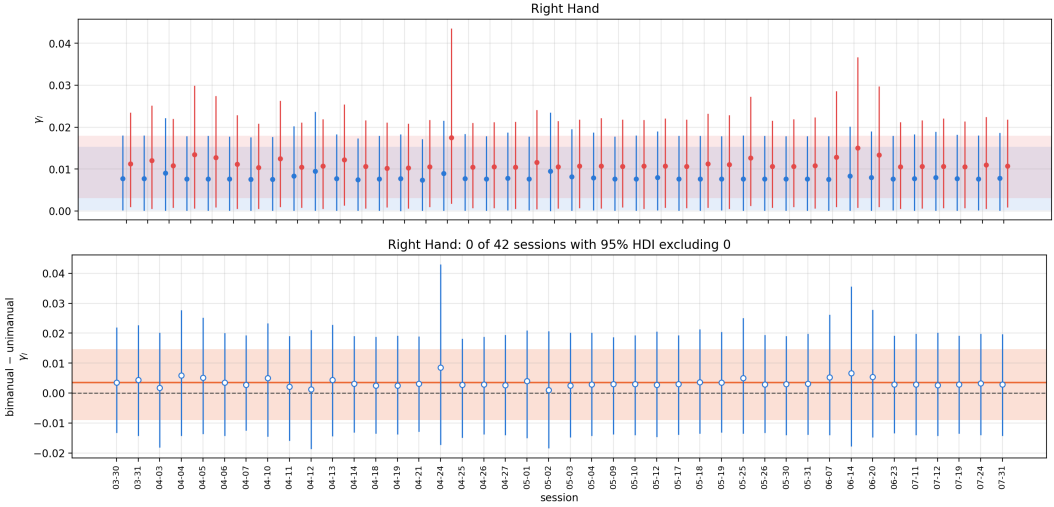
● unimanual, ● bimanual: session mean and 95% HDI; bands: typical session

bi — uni: filled if the session's 95% HDI excludes 0; orange band: average over sessions

Model B on the real data: γ_l, left hand



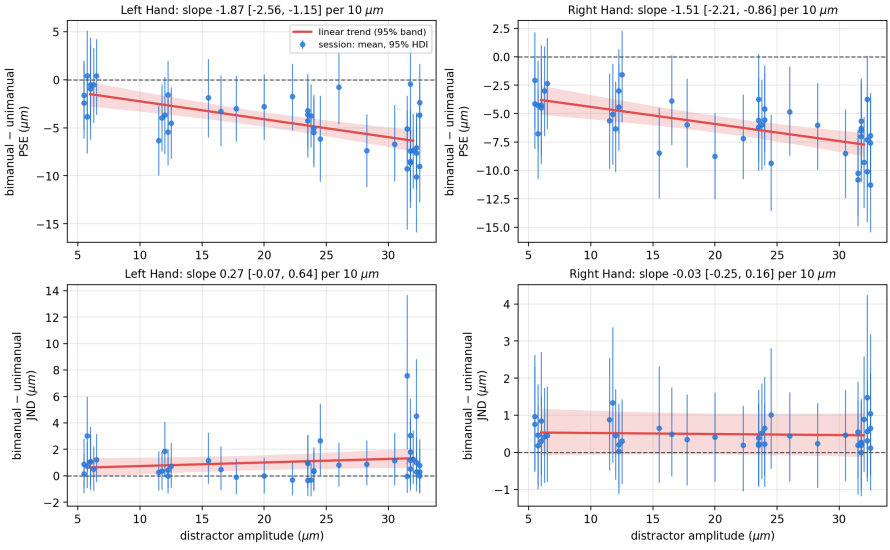
Model B on the real data: γ_l, right hand



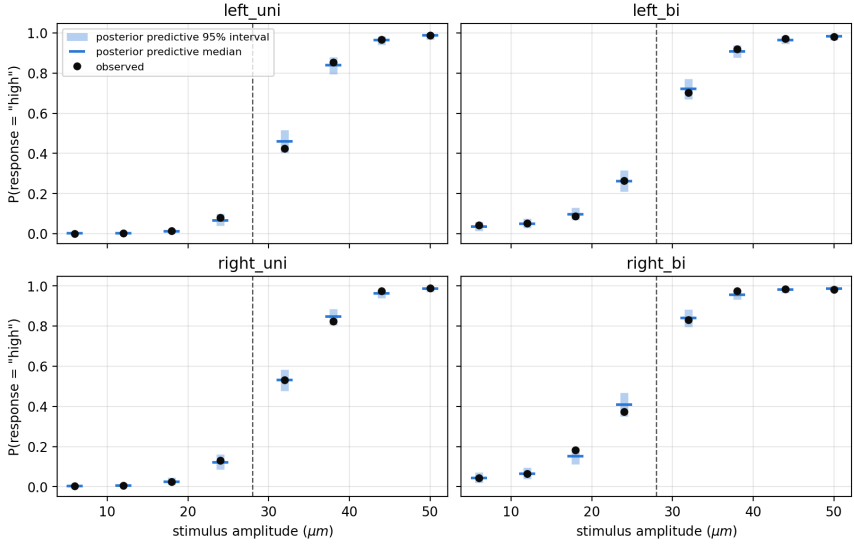
● unimanual, ● bimanual: session mean and 95% HDI; bands: typical session

bi − uni: filled if the session's 95% HDI excludes 0; orange band: average over sessions

Model B: PSE shift vs distractor amplitude (exploratory)



Model B on the real data: the pooled data are reproduced



Model B: conclusions

- Reliable fit; no systematic misfit, pooled or within sessions ($\text{elpd}_{\text{loo}} = -3180.0$, SE 64.3)
- Effect of the distractor (bimanual − unimanual), average over the 42 sessions, posterior mean [95% HDI]:
  - **PSE:** lowered, by -4.29 [-5.05, -3.52] μm (left) and -6.06 [-6.80, -5.34] μm (right); posterior mean negative in 40/42 (left) and 42/42 (right) sessions
  - **JND:** raised for the left hand, $+1.03$ [0.46, 1.59] μm, driven by a few sessions (possibly lapses absorbed into the slope); unclear for the right, $+0.49$ [-0.07, 1.03] μm
  - γ_h**:** raised, by $+0.025$ [0.009, 0.042] (left) and $+0.035$ [0.016, 0.054] (right)
  - γ_l**:** no effect, 0.000 [-0.014, 0.013] (left) and $+0.003$ [-0.009, 0.015] (right)
- The PSE shift grows with distractor amplitude, by about -2 μm per 10 μm (exploratory; amplitude is confounded with session order)

$\Rightarrow$ Put the distractor amplitude into the model: Model C

Model C: including the distractor as a covariate

Psychometric function:

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}\left(\beta_0 + \beta_1 x_{\text{stim}}\right)$$

Suppose the distractor has an impact on the perceived stimulus as follows:

$$x_{\text{perc}} = x_{\text{stim}} + c\, x_{\text{dist}}$$

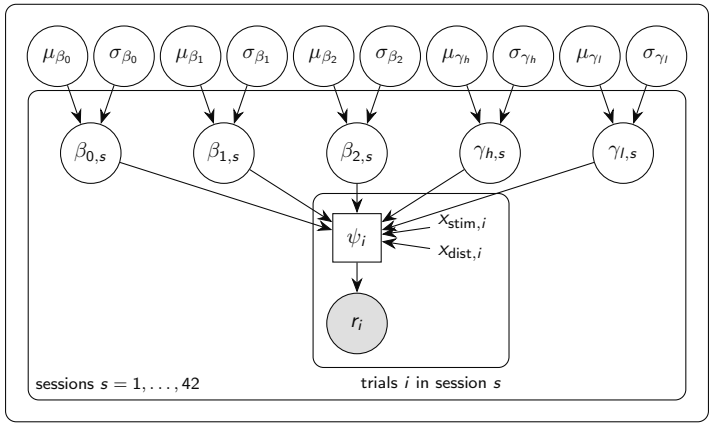
Then:

$$\begin{aligned}\beta_0 + \beta_1 x_{\text{perc}} &= \beta_0 + \beta_1(x_{\text{stim}} + c\, x_{\text{dist}}) \\ &= \beta_0 + \beta_1 x_{\text{stim}} + \beta_1 c\, x_{\text{dist}} \\ &= \beta_0 + \beta_1 x_{\text{stim}} + \beta_2\, x_{\text{dist}}\end{aligned}$$

Considering x_{dist} as a covariate:

$$\psi(x_{\text{stim}}, x_{\text{dist}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}\left(\beta_0 + \beta_1 x_{\text{stim}} + \beta_2\, x_{\text{dist}}\right)$$

Model C: graphical model



Circles: random variables (shaded: observed); rectangle: deterministic; $x_{\text{stim},i}$, $x_{\text{dist},i}$: fixed covariates. The same model is fit independently to each hand, with unimanual and bimanual trials together.

Model C: likelihood

For each hand, fit independently to its $n \approx 6{,}470$ unimanual and bimanual trials, with the distractor as a covariate $x_{\text{dist},i}$ (amplitude / 14.63; $x_{\text{dist},i} = 0$ on unimanual trials):

Likelihood: probability of obtaining responses $\vec{r}$ given parameters $\vec{\theta} = (\vec{\theta}_s)_{s=1}^{42}$,
$\vec{\theta}_s = (\beta_{0,s}, \beta_{1,s}, \beta_{2,s}, \gamma_{h,s}, \gamma_{l,s})$
$r_i \sim \textbf{Bernoulli}[\psi(x_{\text{stim},i}, x_{\text{dist},i}; \vec{\theta}_{s[i]})]$, with
$\psi(x_{\text{stim}}, x_{\text{dist}}; \vec{\theta}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}} + \beta_2 x_{\text{dist}})$, so:

$$\mathbf{P}[r_i = z] = \psi(x_{\text{stim},i}, x_{\text{dist},i}; \vec{\theta}_{s[i]})^z \left(1 - \psi(x_{\text{stim},i}, x_{\text{dist},i}; \vec{\theta}_{s[i]})\right)^{1-z}, \qquad z \in \{0, 1\}$$

$$\begin{aligned} L(\vec{\theta}; \vec{r}) &= \prod_{i=1}^{n} \mathbf{P}[r_i \mid \vec{\theta}_{s[i]}] \\ &= \prod_{s=1}^{42} \prod_{i:\, s[i]=s} \psi(x_{\text{stim},i}, x_{\text{dist},i}; \vec{\theta}_s)^{r_i} \left(1 - \psi(x_{\text{stim},i}, x_{\text{dist},i}; \vec{\theta}_s)\right)^{1-r_i} \end{aligned}$$

Model C: priors

Session parameters drawn from the hand's population (non-centered in the code):

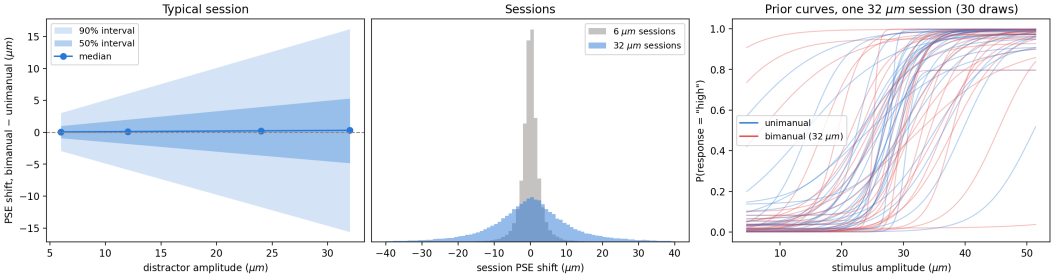
$$\beta_{k,s} \sim \mathcal{N}(\mu_{\beta_k},\ \sigma_{\beta_k}),\ k = 0, 1, 2, \quad \gamma_{h,s} = 0.25\, w_{h,s}, \quad w_{h,s} \sim \text{LogitNormal}(\mu_{\gamma_h},\ \sigma_{\gamma_h})$$
$$\gamma_{l,s} = 0.25\, w_{l,s}, \quad w_{l,s} \sim \text{LogitNormal}(\mu_{\gamma_l},\ \sigma_{\gamma_l})$$

Hyperpriors, the same for both hands; all but those of β_2 as in Model B:

$$\mu_{\beta_0} \sim \mathcal{N}(0,\ 2), \qquad \sigma_{\beta_0} \sim \text{HalfNormal}(2),$$
$$\mu_{\beta_1} \sim \text{LogNormal}(\log 4,\ 0.5), \qquad \sigma_{\beta_1} \sim \text{Exponential}(1),$$
$$\mu_{\beta_2} \sim \mathcal{N}(0,\ 1), \qquad \sigma_{\beta_2} \sim \text{HalfNormal}(1),$$
$$\mu_{\gamma_h},\ \mu_{\gamma_l} \sim \mathcal{N}(-3,\ 1.5), \qquad \sigma_{\gamma_h},\ \sigma_{\gamma_l} \sim \text{HalfNormal}(0.5)$$

- μ_{β_2} centered at 0: the distractor could push responses either way
- Uni- and bimanual trials share slope and lapse rates: $\beta_2 x_{\text{dist}}$ only shifts the curve

Model C: prior predictive check of the distractor effect



Model C drawback: the PSE shift $-\beta_2 x_{\text{dist}}/\beta_1$ is tied to the distractor amplitude, so weak distractors can only shift the PSE slightly

Model C on synthetic data: the study

- True hyperparameters derived from Model B's fit to the real data, so the synthetic data resemble the real data (e.g. $\mu_{\beta_2} = 0.93$ left, 1.13 right)
- Each simulated session gets one distractor amplitude, drawn from the real sessions' amplitudes
- The same two sweeps as for Model B, each run twice with new data:
  - sessions: 5, 10, 20, 40, 80 sessions of 308 trials
  - trials: 42 sessions of 20, 40, 80, 160, 320 trials
- 20 fits: 2 divergences in total (in the two smallest datasets), $\hat{R} \leq 1.008$, bulk ESS ≥ 532

Model C on synthetic data: hand-level means

At 40 sessions: true value and posterior mean [95% HDI], averaged over the two replicates

	left		right	
	true	posterior	true	posterior
μ_{β_0}	-1.49	-1.56 [-1.90, -1.24]	-1.06	-0.94 [-1.29, -0.57]
μ_{β_1}	4.64	4.65 [4.16, 5.18]	4.46	4.46 [4.06, 4.86]
μ_{β_2}	0.93	0.93 [0.71, 1.18]	1.13	1.11 [0.74, 1.47]
μ_{γ_h}	-2.80	-2.89 [-3.45, -2.41]	-2.41	-2.60 [-3.10, -2.11]
μ_{γ_l}	-3.20	-3.39 [-4.34, -2.60]	-3.20	-3.04 [-3.73, -2.37]

- All five recovered without visible bias; μ_{β_2}'s HDI excludes 0
- Lapse means (logit scale) are the least precise

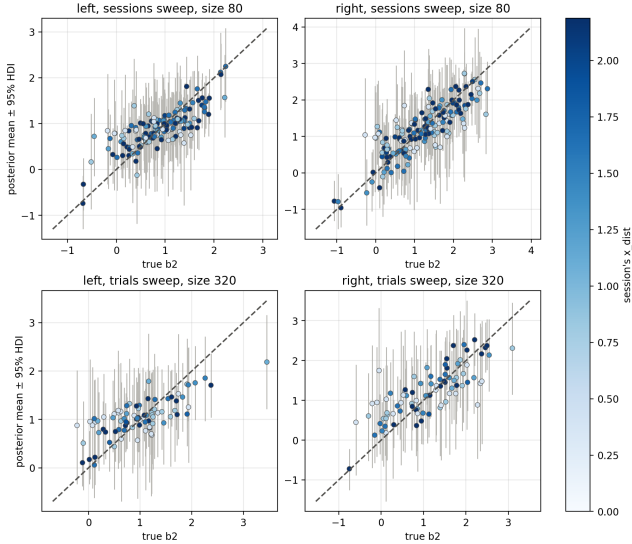
Model C on synthetic data: between-session sds

At 80 sessions: true value and posterior mean [95% HDI], averaged over the two replicates

	left		right	
	true	posterior	true	posterior
σ_{β_0}	0.76	0.74 [0.58, 0.92]	1.01	0.94 [0.76, 1.14]
σ_{β_1}	0.78	0.83 [0.57, 1.09]	0.39	0.17 [0.00, 0.40]
σ_{β_2}	0.62	0.51 [0.36, 0.67]	0.83	0.86 [0.68, 1.06]
σ_{γ_h}	0.48	0.38 [0.00, 0.83]	0.37	0.31 [0.00, 0.68]
σ_{γ_l}	0.38	0.35 [0.00, 0.85]	0.41	0.29 [0.00, 0.69]

- The sds of β_0, β_1, β_2 are identified (a small σ_{β_1} is hard to tell from 0)
- The lapse sds are not: their HDIs start at 0 and barely narrow with more data

Model C on synthetic data: session β_2 is weak for small distractors



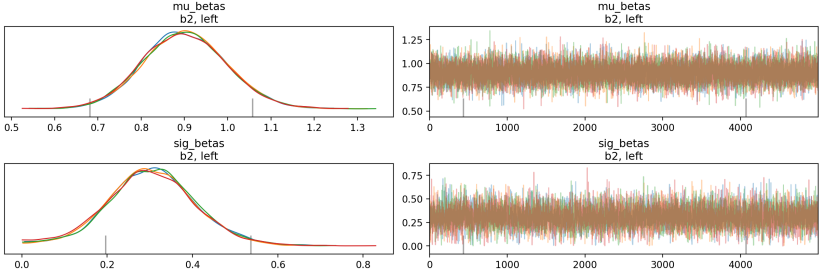
Model C on synthetic data: what is recoverable

At the size of the real data (42 sessions of about 308 trials), if the data follow Model C:

- **Recoverable:** per-session PSEs in both conditions and JNDs; all five hand-level means, including the distractor effect μ_{β_2}; the between-session sds of β_0, β_1 and β_2
- **Weakly determined:** the individual session coefficients, especially β_2 in sessions with a small distractor (HDI 2.6 wide at 6 μm, 1.2 at 32 μm), which are pulled toward the hand mean
- **Not identified:** σ_{γ_h} and σ_{γ_l}; their posteriors reflect the prior
- **Not tested:** what happens when the data break Model C's assumptions (shared lapse rates and slope, a shift proportional to amplitude)

Model C on the real data: sampling diagnostics

- NUTS, 4 chains $\times$ 5,000 draws after 1,000 tuning (9 min)
- 2 divergences in 20,000 draws, in no particular region; tree depth never above 5
- All 1,300 variables: $\hat{R} = 1.000$, bulk ESS $\geq$ 3,410, tail ESS $\geq$ 5,231
- HalfNormal sds defined directly, so no sign warning as in Model B
- Below: the left hand's distractor hyperparameters μ_{β_2}, σ_{β_2}



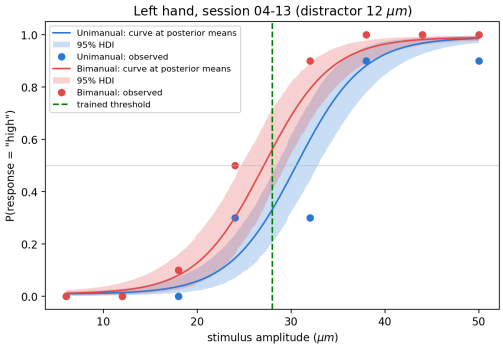
Model C on the real data: hand-level estimates

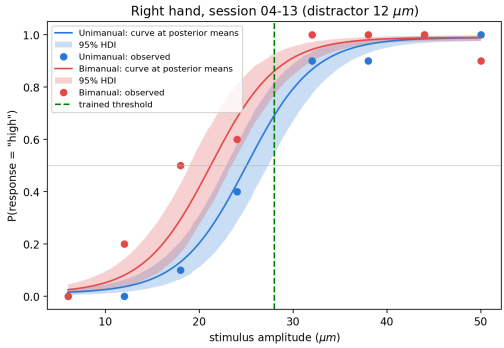
	left	right
distractor coefficient μ_{β_2}	0.90 [0.71, 1.09]	1.18 [0.99, 1.37]
PSE, unimanual (μm)	32.34 [31.51, 33.17]	31.15 [29.92, 32.40]
JND (μm)	3.75 [3.39, 4.11]	3.88 [3.59, 4.19]
γ_h	0.0091 [0.0033, 0.0147]	0.0113 [0.0042, 0.0189]
γ_l	0.0086 [0.0023, 0.0155]	0.0085 [0.0029, 0.0143]
σ_{β_0} / σ_{β_1} / σ_{β_2}	0.60 / 0.90 / 0.30	1.07 / 0.35 / 0.23

Posterior mean [95% HDI]; PSE, JND and lapse rates of a typical session (JND and lapse rates are the same on bimanual trials).

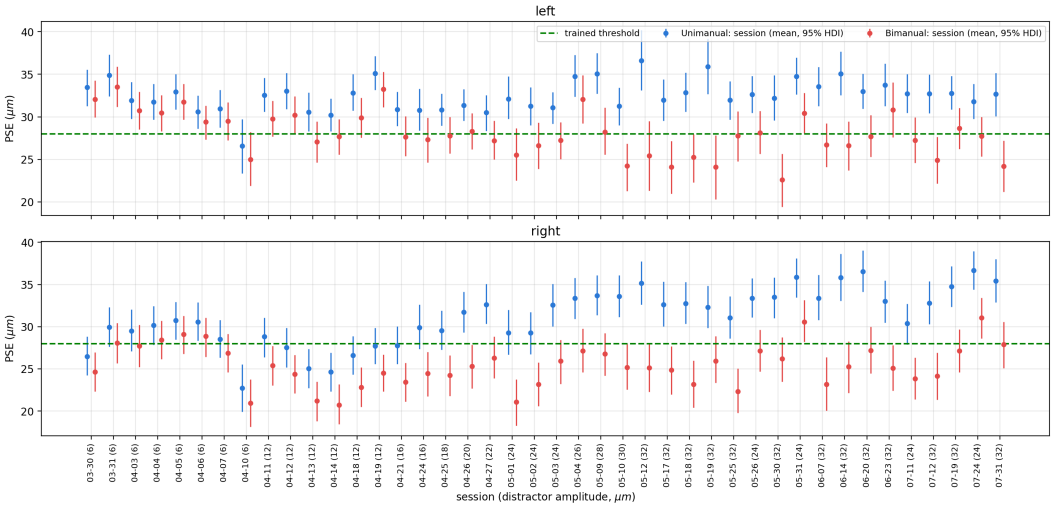
- $\mu_{\beta_2} > 0$ for both hands: the distractor lowers the PSE, more the stronger it is
- The sessions vary little around the hand's amplitude line (σ_{β_2} small)

Model C on the real data: fitted curves in one session (04-13)

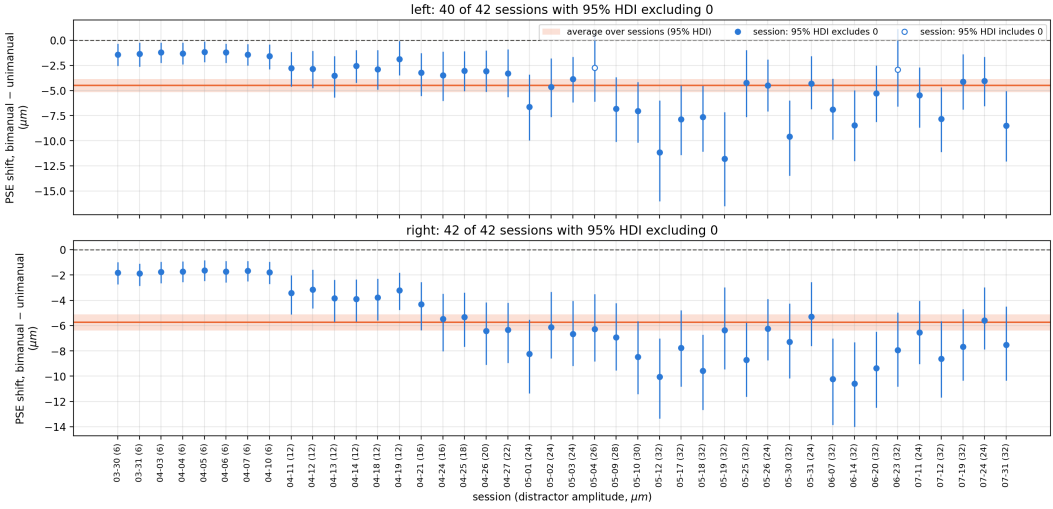




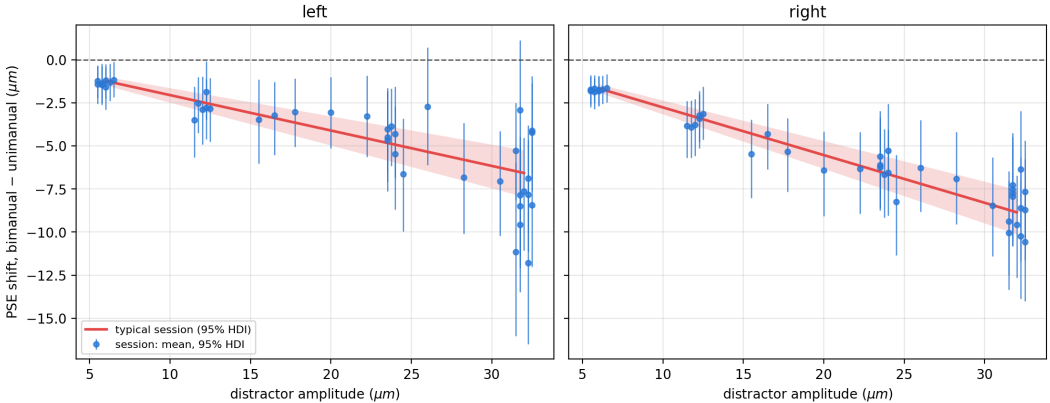
Model C on the real data: PSE in each session



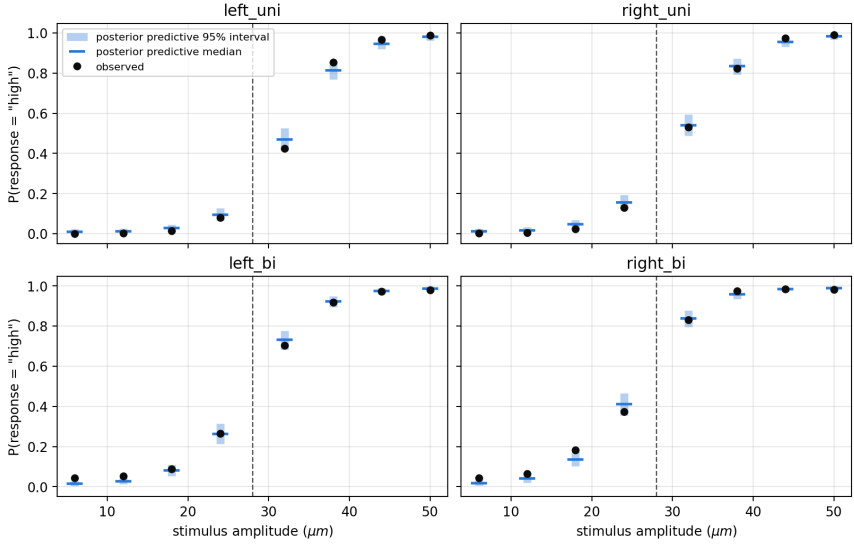
Model C on the real data: effect of the distractor on PSE



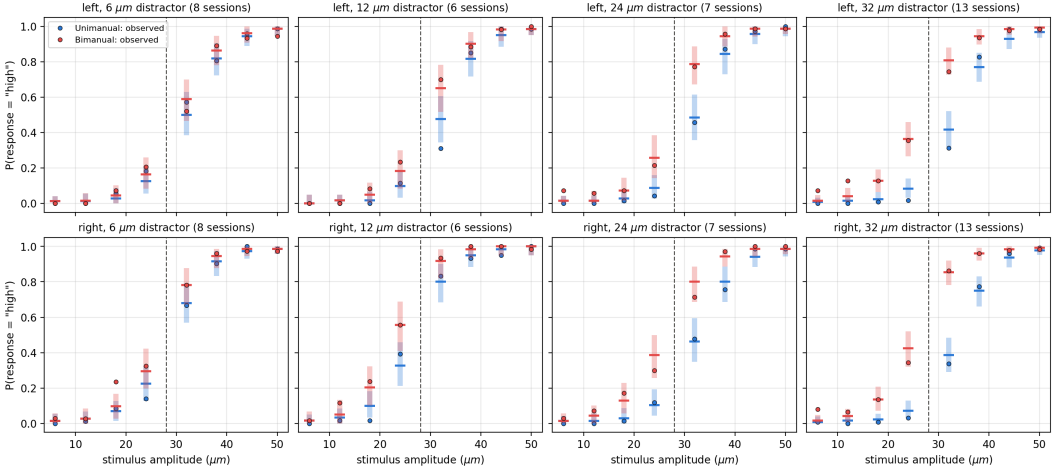
Model C: the PSE shift grows with distractor amplitude



Model C on the real data: misfit at the weakest stimuli



Model C: observed vs predicted, by distractor amplitude



Model C: conclusions

- Reliable fit: 2 divergences in 20,000 draws, $\hat{R} = 1.000$ ($\text{elpd}_{\text{loo}} = -3217.0$, SE 65.7)
- Effect of the distractor (bimanual − unimanual), posterior mean [95% HDI]:
  - **PSE:** lowered, by -4.51 [-5.15, -3.86] μm (left) and -5.74 [-6.39, -5.11] μm (right) on average over sessions; more the stronger the distractor: a typical session shifts -1.23 and -1.66 μm at 6 μm, -6.57 and -8.85 μm at 32 μm
  - **JND, γ_h, γ_l:** shared by both conditions, so the model cannot show an effect on them
- The misfit sits where its assumptions meet the data:
  - shared lapse rates: 6 of 32 pooled proportions outside their intervals, all at weak stimuli
  - a shift proportional to amplitude: for the right hand, even a 6 μm distractor shifts responses more than Model C allows

$\Rightarrow$ Which model describes the data best? Model comparison

Leave-one-out cross-validation

All three models fit to the same 12,937 trials in the same order, so the difference between two models can be computed trial by trial:

Model	elpd_{loo} (SE)	p_{loo}	difference (SE)
A: sessions pooled	-3344.0 (64.5)	13.4	B $-$ A: $+164.1$ (19.2)
B: sessions hierarchical	-3180.0 (64.3)	178.7	**best**
C: distractor as covariate	-3217.0 (65.7)	138.6	C $-$ B: -37.1 (11.9)

- B beats A by 8.5 standard errors, despite 179 effective parameters against 13
- B beats C by 3.1 standard errors, although C needs fewer effective parameters
- Every Pareto k below 0.7, so the PSIS-LOO estimates are reliable

Model recovery: is B's advantage real?

Two questions before reading B's win as session-to-session variation:

1. Can LOO identify the model that generated a dataset of this size and design?
2. Could B win just because it is more flexible? B contains A as the case of no between-session variation.

Design:

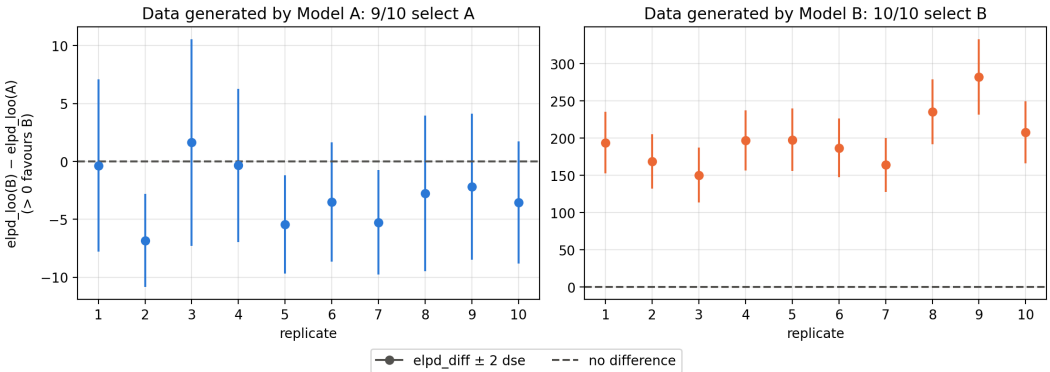
- 10 replicates, each with two synthetic datasets on the real trial design (12,937 trials in 42 sessions)
  - **data A:** responses from Model A, true values = A's real-data posterior means
  - **data B:** session parameters drawn from B's real-data hyperparameters
- Both models fit to every dataset; LOO picks the model with the higher elpd_{loo}; $\Delta = \text{elpd(B)} - \text{elpd(A)}$

Model recovery: LOO picks the generating model

	selects A	selects B
data from A	9	1
data from B	0	10

- **Data from B:** B wins every time by a wide margin, $\Delta = 150.2$ to 282.4 ($z = 8.2$ to 11.1); B's $p_{\text{loo}} = 167$–190
- **Data from A:** A wins 9 of 10, but narrowly, $\Delta = -6.8$ to $+1.6$; the one B win is a near-tie ($z = 0.36$). B mimics A by shrinking its sds ($p_{\text{loo}} = 54$–84)
- **Real data:** $\Delta = 164.1$ and $p_{\text{loo}} = 178.7$, inside the data-from-B range and far outside the data-from-A range

Model recovery: Δ in every replicate



Model recovery: conclusions

- With this design and data size, LOO identifies session-to-session variation reliably: B wins every time with $z > 8$ when the variation is there
- B's flexibility alone does not make it win: on data without session variation it loses or ties ($\Delta \leq 1.6$)
- The real data look like the data-from-B replicates in both Δ and p_{loo}: the psychometric curves do vary between sessions
- The reverse, confirming that variation is absent, is only weakly answered: the more flexible model loses by just a few elpd points when the simpler one is true
- Limits: 10 replicates per model; one set of B hyperparameters (with smaller between-session sds the gap would shrink)

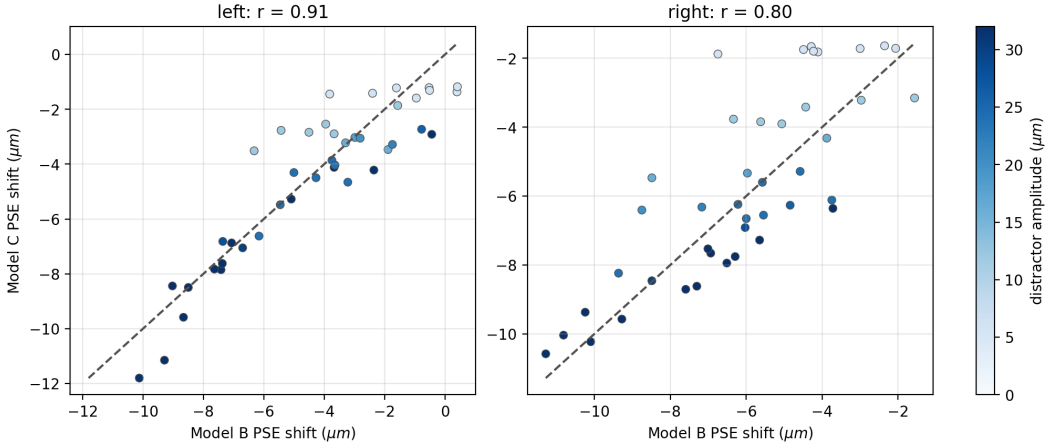
Model C vs Model B: where C loses

$\text{elpd}_{\text{loo}}(\text{C}) - \text{elpd}_{\text{loo}}(\text{B})$ summed over the trials at each amplitude; negative values favor Model B

	left		right	
stimulus (μm)	bimanual	unimanual	bimanual	unimanual
6	−6.6	−3.5	−5.1	−2.3
12	−4.2	−3.0	−1.8	−2.8
18	0.7	−1.6	−3.4	−0.3
24	−1.0	0.8	1.9	0.8
32	−0.1	−3.4	0.3	1.7
38	−0.3	−1.4	0.2	1.6
44	0.0	−3.8	0.4	−1.5
50	−0.1	0.3	0.1	0.0

- 29.2 of the 37.1 are lost at 6 and 12 μm, on unimanual as well as bimanual trials: one lapse rate for both conditions, the main cause
- Next: left unimanual at 32 and 44 μm: one slope for both conditions

Model C vs Model B: session PSE shifts



Model C vs Model B: the weakest distractor

Session PSE shifts (μm): mean, and range of the posterior means, over the sessions at the weakest and strongest distractor

hand	distractor	Model B	Model C
left	6 μm (8 sessions)	-1.13 (-3.82 to 0.41)	-1.34 (-1.59 to -1.18)
left	32 μm (13 sessions)	-6.67 (-10.12 to -0.44)	-7.40 (-11.80 to -2.91)
right	6 μm (8 sessions)	-3.91 (-6.74 to -2.06)	-1.75 (-1.88 to -1.64)
right	32 μm (13 sessions)	-7.90 (-11.28 to -3.72)	-8.58 (-10.58 to -6.36)

- The models agree on the average shift and, for the left hand, on the session shifts ($r = 0.91$)
- For the right hand at 6 μm, B's shift is more than twice C's: C ties the shift to the amplitude, so a weak distractor must give a small shift
- Hints at an effect of the distractor being present, on top of its amplitude (right hand); not yet shown to improve prediction

Model comparison: conclusions

- **Sessions differ:** B predicts far better than A, and model recovery shows this is not B's flexibility alone.
- **The models agree on the PSE shift:** about -4.2 to -4.5 μm (left) and -5.7 to -6.1 μm (right) on average.
- **The shift depends on amplitude:** C confirms the trend inside a model.
- **B predicts better mainly because of lapses:** C's one lapse rate for both conditions misses the extra "high" responses to weak stimuli on bimanual trials. C's misfit points to
  - lapse rates per condition (the strongest evidence); open: does the bimanual γ_h grow with distractor amplitude?
  - less clearly, a distractor-presence term (right hand) and a bimanual slope

$\Rightarrow$ Model B is the model to use for inference for now

Comparison of the models

	A	B	C
Sessions	pooled	hierarchical	hierarchical
Distractor	separate groups	separate groups	covariate $\beta_2 x_{\text{dist}}$
elpd_{loo} (SE)	-3344.0 (64.5)	-3180.0 (64.3)	-3217.0 (65.7)
difference from B (SE)	-164.1 (19.2)	best	-37.1 (11.9)
effective parameters p_{loo}	13.4	178.7	138.6
PSE shift, left / right (μm)	-4.19 / -6.07	-4.29 / -6.06	-4.51 / -5.74

PSE shift: bimanual − unimanual, posterior mean; for B and C averaged over the 42 sessions. Numbers from Fit_A, Fit_B and Fit_C_Report.

- All three models agree on the PSE shift, to within about 0.3 μm for each hand
- B predicts best: sessions differ (B $\gg$ A), and lapse rates differ between conditions (B $>$ C)

Conclusions

- The distractor lowers the PSE: by about 4.2–4.5 μm (left) and 5.7–6.1 μm (right), in all three models.
- It raises the rate of "high" responses to weak stimuli on bimanual trials (γ_h); whether this grows with distractor amplitude is open.
- The sessions differ; a hierarchical model is needed.
- The shift grows with distractor amplitude, and a weak distractor may shift the PSE more than a proportional model allows (right hand).

Next steps

- A new Model D, revising Model C: lapse rates per condition, testing whether the bimanual γ_h depends on distractor amplitude (or session order).
- A distractor-presence term only if it improves prediction.
- Model recovery for B against D.